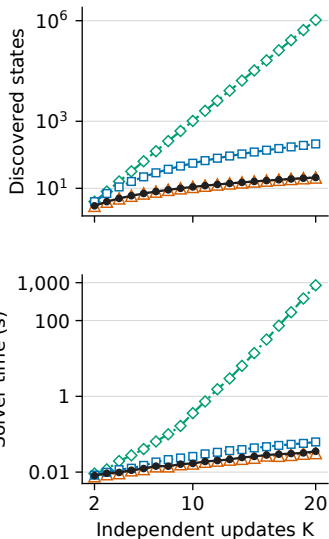


(a) Independent

19 K values, four methods



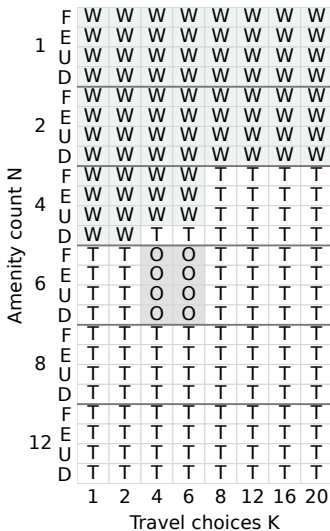
—●— F: FG lazy    —□— E: Eager    —△— U: Update-first    —◇— D: Direct-Full

Hub:

—○— No UC    —□— Local binding    —△— Cross binding

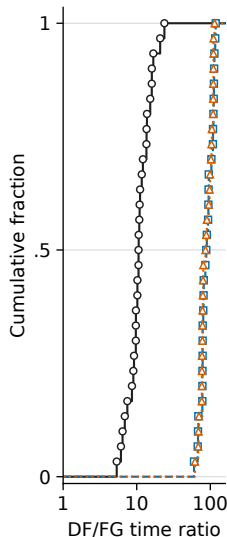
(b) Travel

48 settings x four methods



(c) Hub

30 pairs/profile



Time: five-run medians; bars: min-max. Hub: all 90 same-instance pairs.

Travel: W = WIN; T = 1,200 s timeout; O = OOM (64 GiB heap).